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Transport properties of imidazolium-based room temperature ionic liquids in confinement of slit charged carbon nanopores: New insights from molecular simulations

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ABSTRACT

The paper presents a study of the behavior of room-temperature ionic liquids with different alkyl chain lengths and anion types in slit negatively charged carbon nanopores of various widths (1–15 nm) using all-atom molecular dynamics simulations. The structure of the confined ionic liquids is investigated using number, mass, and charge density profiles and radial distribution functions. The paper also discusses such characteristics as mean squared displacement of ions in all directions, self-diffusion coefficients of cations and anions in the bulk, their temperature dependences, electrical conductivity. Some energy characteristics are obtained by quantum chemical calculations. It is established that there is a bulk region in the pore center for all the confined ionic liquids, starting from the pore width of 7 nm. The sums of the cationic and anionic self-diffusion coefficients of different ionic liquids can be arranged as follows: D[OMIM][BF_4] < D[EMIM][BF_4] < D[OMIM][NTf_2] < D[EMIM][NTf_2]. Due to layering structure inside the pore, the dynamics of the ions is heterogeneous. It is shown that the RTILs with longer alkyl chains have lower electrical conductivities regardless of temperature. The opposite effect associated with the species size is observed for RTILs with different anions.

1. Introduction

Room temperature ionic liquids (RTILs) are among the most attractive materials for numerous fields of science and engineering. Their unique physicochemical properties, such as negligible vapor pressure, high viscosity and ion density, nonflammability, excellent thermal, chemical, and electrochemical stability, and wide electrochemical window make them promising for a number of applications, including electrochemistry, heterogeneous catalysis, biocatalysis, extraction, nanoparticle synthesis, optics, drug delivery, sensing, and biosensing [1–10]. RTILs have been proposed as an alternative to conventional electrolytes in electrochemical double-layer capacitors (EDLCs) [11–14] and in dye-sensitized solar cells (DSSCs) for solar energy conversion [15–16]. Commercial interest in such an important technology, as electrochemical energy storage, provokes the interest of the scientific community in ILs confined inside materials with nanometer-sized pores. Information about the behavior of RTILs in confinement, in particular their structural, dynamic, and electrical properties inside nanopores, is crucial to the rational design of EDLCs and DSSCs with the optimal properties. Computational methods, such as molecular dynamics, along with experimental techniques can help better understand the properties of RTILs inside nanopores and, in some cases, obtain detailed information at the molecular level, which is experimentally inaccessible. A number of works were devoted to studying the behavior of RTILs confined inside pores of various geometries, including slit pores [17–21] and cylindrical pores [22–24]. In addition to the specific information that these works provide, they also contain a number of general conclusions. In agreement with each other the authors state that the structure and dynamics of confined RTILs are affected by the confinement, surface interactions, and reduced dimension of the system. In particular, in work [21] the authors investigate the structure and dynamics of 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([BMIM][NTf_2]) confined at ambient temperature and pressure in hydroxylated amorphous silica nanopores and show that the confined

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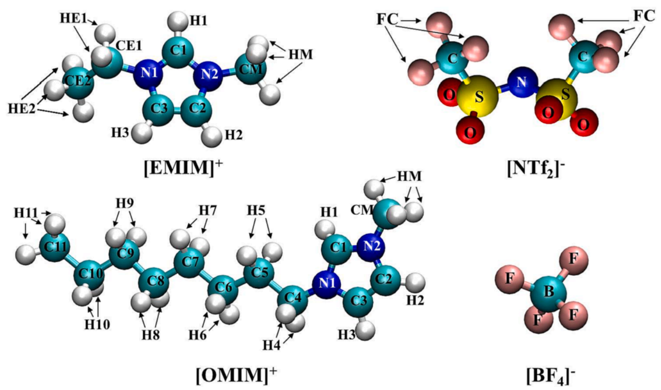


Fig. 1. Geometrically optimized structure of the cations ($[EMIM]^+$ and $[OMIM]^+$) and anions ($[NTf_2]^-$ and $[BF_4]^-$).

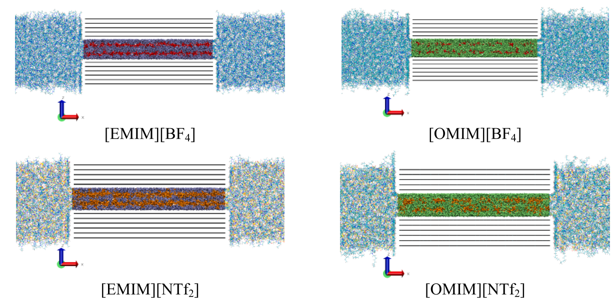


Fig. 2. Snapshots of the part of the simulation boxes for the systems with h = 2 nm. In the pore region the ice blue and green colors represent the $[EMIM]^+$ and $[OMIM]^+$ cations, the red and orange ones represent the $[BF_4]^-$and $[NTf_2]^-$ anions. Black represents the carbon atoms of the graphene sheets.

RTIL forms well-defined layers on the pore surface and the structural and dynamic properties of $[BMIM][NTf_2]$ can be described as the sum of the surface contribution arising from the ions adsorbed on the surface and the bulk-like contribution arising from the ions located in the pore center. For $[BMIM][PF_6]$ [23], $[EMIM][NTf_2]$ [19,20], and $[EMIM][PF_6]$ [25] it is demonstrated that, besides the pore size, the properties of the confined RTILs are greatly affected by pore loading.

Notably, the dynamic properties of confined RTILs still remain unclear to a certain extent and are a matter of debate. The first experimental observation that cations of RTIL, namely $[BMIM][NTf_2]$, that do not become immobilized near the pore walls exhibit higher diffusivity than in the bulk liquid was reported by S. Chathoth and coauthors [26]. At the same time, using the NMR method and MD simulations Li et al. observed a slowdown in ion diffusion ($[BMIM][NTf_2]$) inside silica and carbon mesopores compared with bulk values [27]. But for $[EMIM][NTf_2]$ confined inside uncharged slit-like graphitic nanopores, a local increase in the translational mobility was observed [19]. Ghoufi A. et al. [28], who studied the local dynamics of three RTILs inside carbon nanotubes (CNT) of different diameters in slit carbon and in cylindrical silica nanopores, reported ultrafast diffusion of RTILs confined in CNT and a slowdown in ion diffusion in silica and slit nanopores compared with bulk. In a recent work [29], the cation dynamics of 1-N-butylpyridinium bis(trifluoromethylsulfonyl)imide in a nanoconfinement of porous carbons with different pore sizes as a function of temperature was investigated using quasi elastic neutron scattering techniques. In contrast to RTILs obtained from a slit-pore confinement [30], the authors [29] showed that the ionic liquid in the nanoporous carbon melts already well below the bulk melting point similarly as is known for many conventional liquids.

Thus, despite numerous experimental and theoretical studies of the diffusion of RTILs in confinement, a complete fundamental understanding of this phenomenon has not yet been achieved. In contrast to the works mentioned above, in the current work we apply all-atom MD to investigate the behavior of RTILs in charged carbon nanopores. In addition to the diffusion of ionic liquids, which differ in the side chain length of the cation and the size of the anion, we consider the structure of RTILs in pores. In addition, the influence of temperature on the behavior of RTILs in pores as well as in bulk is discussed.

2. Simulation details

2.1. MD calculations

The behavior of four ionic liquids confined in a charged slit carbon nanopore was investigated by means of classical molecular dynamics (MD) in the NVT-ensemble with GROMACS package version 2019.6 [31]. The calculations were performed on the supercomputer facilities provided by NRU HSE [32]. We chose 1-ethyl-3-methylimidazolium $[EMIM]^+$ and 1-octyl-3-methylimidazolium $[OMIM]^+$ cations and tetrafluoroborate $[BF_4]^-$ and bis(trifluoromethylsulfonyl)imide $[NTf_2]^-$ anions (Fig. 1) for the MD simulation. The potential model which best reproduces the structural, thermodynamic and dynamic properties [33] was used for all the ionic liquids with the charge scaling factor = 0.8.

Each RTIL was placed into a rectangular box consisting of a slit carbon nanopore in the center and two reservoirs on its left and right sides. Detailed information about the construction of the simulation box and making of the slit charged nanopore consisting of graphene sheets was given in our previous work [34]. The innermost graphene sheets were modeled as a cathode by uniformly distributing negative partial charges throughout the pore, with the charge density equal to −0.8 e/nm^2 (-12.8 μC/cm^2), where e is the elementary charge. The interactions between IL and the wall were represented by a 6–12 Lennard-Jones potential. The parameters for the carbon atoms were set at ε = 0.29288 kJ/mol and σ = 0.355 nm [35]. To calculate the cross interactions, the combination rule was employed: $\sigma_{ij} = (\sigma_i\sigma_j)^{1/2}$ and $\varepsilon_{ij} = (\varepsilon_i\varepsilon_j)^{1/2}$. The number of RTIL ions was initially set to correspond to the density equal to the bulk density at 300 K and 0.1 MPa. The densities of all the RTILs at ambient conditions as well as at other temperatures were taken from literature data [36–38]. To ensure that these systems were electrically neutral, we removed the excess anions. Periodic boundary conditions were applied in all the three directions. Tables 1S-4S in Supplementary Material present the systems composition. Fig. 2 shows, as an example, snapshots of the part of the simulation boxes for the systems with the pore width of h = 2 nm.

The systems were energy-minimized using the steepest descent energy minimization algorithm. The temperature was maintained using a Nose–Hoover thermostat with a 0.1 ps coupling constant [39,40]. The leapfrog algorithm was used to integrate the motion equation. The integration step was fixed at 1 fs. The Coulomb long-range interactions were studied by the Particle Mesh Ewald (PME) method with an interpolation order of 4 and 0.25 nm grid spacing. The cutoff distance for all the interactions was set to 1.2 nm. In all of the simulations, the h-bond length was constrained by the LINCS algorithm. In our previous work [34] we showed that a 40 ns production run was enough to achieve a constant density inside the pore and to make sure that the results were statistically converged. Fig. 3a presents the number of cations and anions of RTILs inside the pore by the end of the simulation. As can be seen, it is a linear function of slit pore width for all the RTILs. As expected, the largest number of RTIL ions consisting of a cation with a smaller alkyl chain and a smaller anion, i.e. $[EMIM][BF_4]$, is located inside a pore of a certain width, and vice versa, the number of $[OMIM][NTf_2]$ in the pore is smaller than for the other RTILs. Fig. 3b demonstrates the dependence of final density of the RTIL inside the pore, ρ_p, on the pore width. The density of the ILs is lower within nanopores compared to the bulk phase due to the confinement effect. When the ILs are confined within nanopores, the ions interact strongly with the charged pore walls, causing a rearrangement of the ionic structure. Our previous work [34] has shown

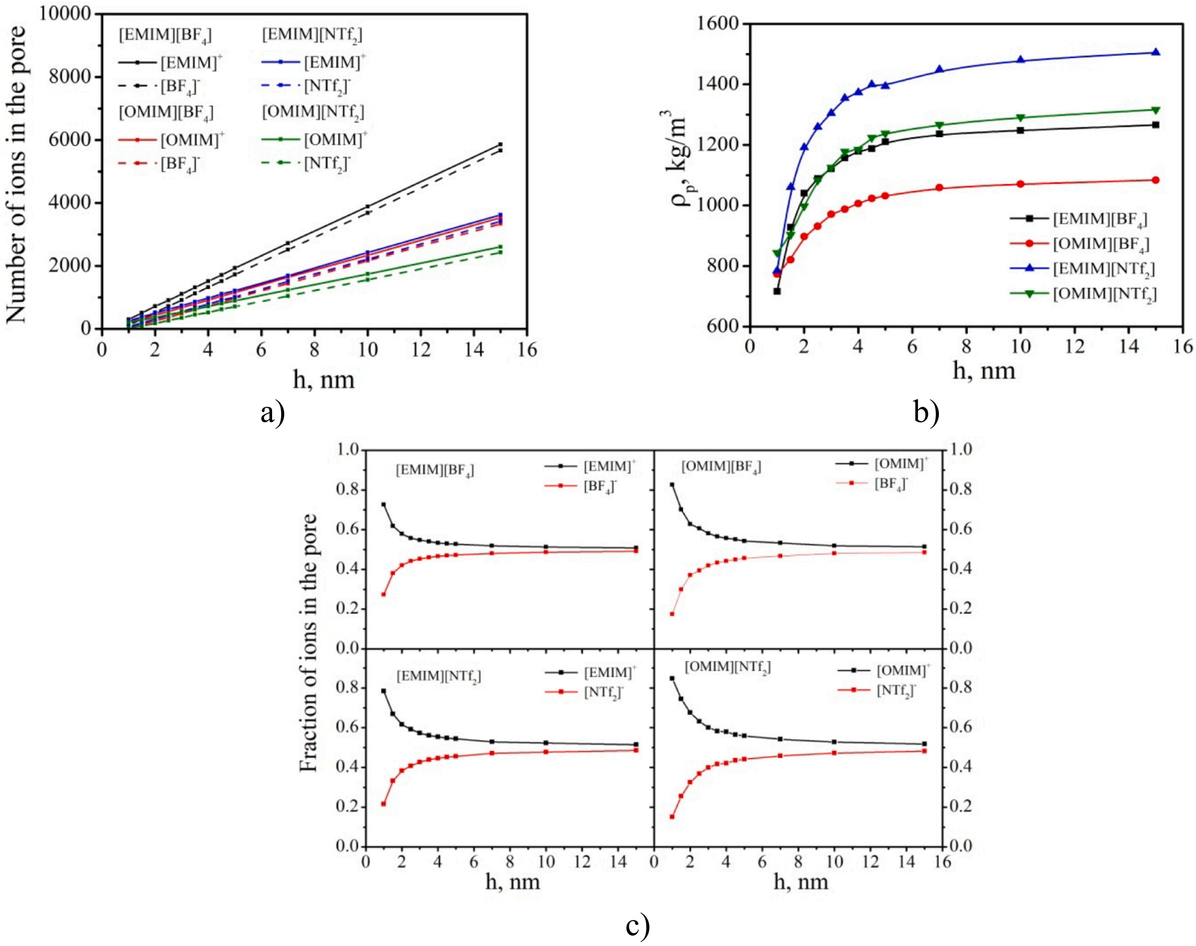


Fig. 3. Number of ions (a), density (b), and fraction (c) of cations and anions of RTILs in the pore as a function of slit pore width, h.

that the structure of the ionic liquid is highly layered at pore widths up to 5 nm. In the bulk phase, the close packing of ions results in a relatively high density. However, within nanopores, the limited space for ion arrangement leads to a decrease in overall density. Additionally, the confinement effect affects the spatial distribution of ions, resulting in more ordered and differently oriented ions near the pore walls compared to the bulk. This further contributes to a decrease in density. However, as the width of the bulk-like layer in the central part of the pore increases, the density of the confined IL also rises. For pores with a width of 15 nm, we observe a density of confined ILs close to that of the bulk ILs. For the RTILs with an $[OMIM]^+$ cation, the $\rho_p(h)$ dependence for h = 1–1.5 nm is seen to deviate from the exponential growth curve/graph. This is due to the fact that the excess charge on the graphene walls can only be compensated for if the pores with $\sigma = -0.8$ e/nm^2 contain at least 200 cations, regardless of their size. In addition, the pores must also have a small fraction of anions to compensate for the strong electrostatic repulsion that occurs between the cations. Fig. 3c shows the fraction of ions inside the pores. In the region of narrow pores, the difference between the fractions of cations and anions is maximal for $[OMIM][NTf_2]$ and is minimal for $[EMIM][BF_4]$, which is associated with the ion size. But with an increase in the pore width, this difference disappears and the ratio of the cation and anion fractions reaches the value of 50/50 for all the RTILs.

2.2. Quantum chemical calculations

All the calculations were carried out in Gaussian 09 [41] package by the density functional theory (DFT) method with the B3LYP [42] functional using Grimme dispersion correction GD3 [43] and the 6–311++g (2d,2p) [44,45] basis set including polarization and diffuse functions. Individual ions and ionic pairs were optimized, the stability of the structures was confirmed by calculating the frequencies. The atomic charges for individual ions were calculated by several methods: HLY [46], MK [47,48], and CHELPG [49]. Also, the electrostatic potential-mapped electronic density surface was calculated. The electron density surface with the density isovalue of 0.001 was considered to correspond to the VdW surface.

The interaction energies for the ionic pairs were calculated taking into account the basis set superposition error (BSSE) [50] by the counterpoise method and energy decomposition analysis was carried out in the Multiwfn package [51]. Also, the interactions were analyzed applying Weinhold's Natural bond orbital (NBO) analysis [52,53] and Bader's Quantum theory of atoms in molecules (QTAIM) [54]. The NBO method was used to calculate the stabilization energy E^2 and charge transfer q values, which characterize the charge transfer from the H-acceptor lone pair (Y) to the non-bonded H-X orbital (LP Y → BD* X-H). The QTAIM method was applied to calculate the electron density ρ and potential energy density V at critical points (BCP) on the respective bond paths. The interaction energy for the respective bond path can be estimated from Espinosa et al.'s correlation [55].

For energy decomposition analysis (EDA), the dispersion energy E_{disp} of the ionic pairs was determined as follows:

$$E_{disp} = E_{int}(B3LYP + GD3)\text{-}E_{int}(B3LYP)$$

assuming the B3LYP functional has no dispersion attraction. The electrostatic energy E_{el} was calculated according to Shubin Liu's energy decomposition [56] for the ionic pair and cation and anion. Then the electrostatic E_{el} term of the interaction energy was calculated as the

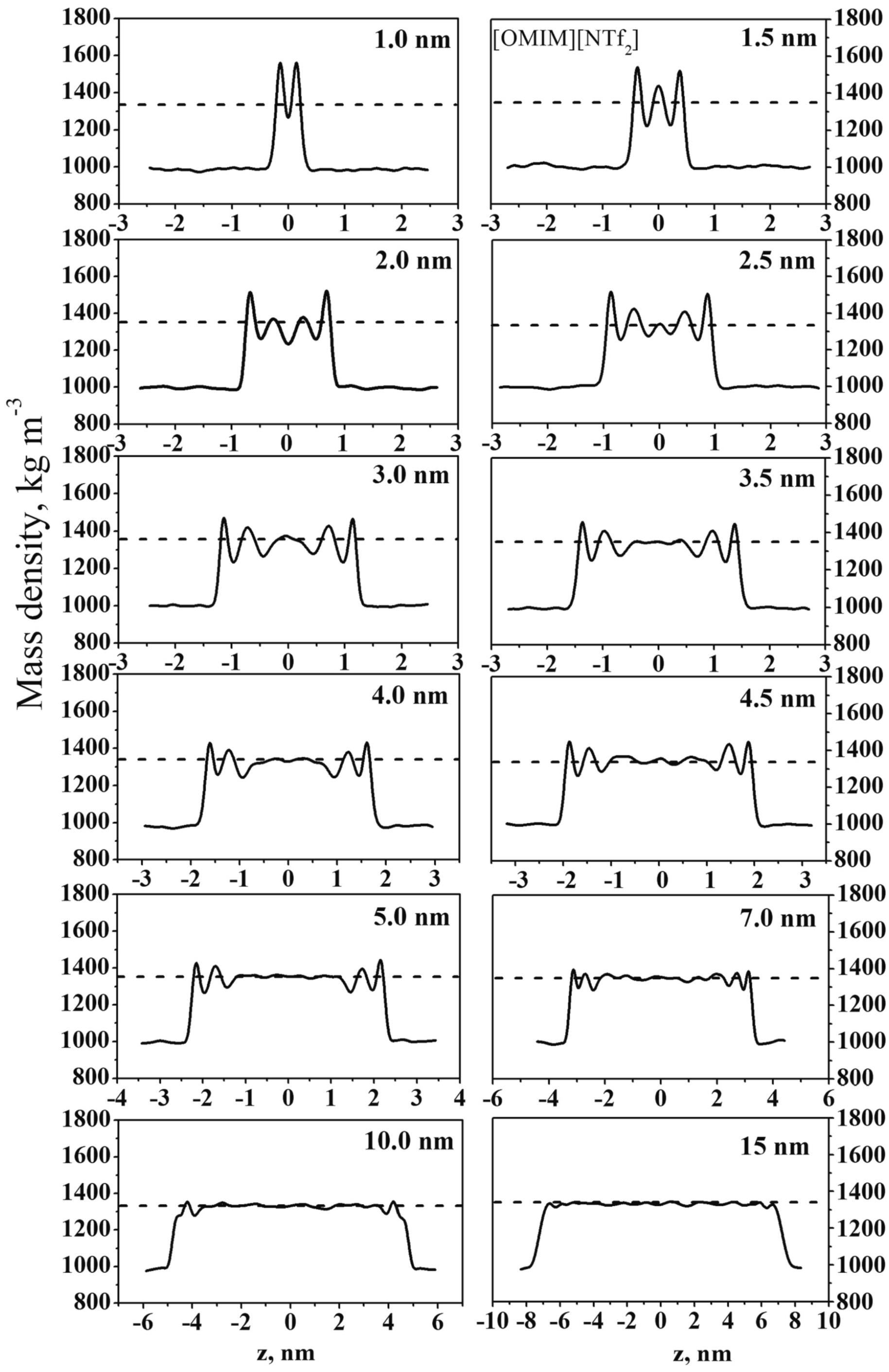


Fig. 4. Mass density distribution profiles of $[OMIM][NTf_2]$ in the direction perpendicular to the graphene surfaces at 300 K for h = 1–15 nm. The average is taken over the whole box. The dashed line denotes the bulk density in the reservoirs.

difference of these energies for the ionic pair and individual ions in the complex. And, finally, the repulsion energy E_{rep} was calculated as the difference between the ionic pair interaction energy E_{int} and the sum of the dispersion E_{disp} and electrostatic E_{el} energies.

3. Results and discussion

3.1. Structural properties

Before discussing the dynamic properties, it is important to obtain information about the structural features of ionic liquids in the limited space of a charged carbon nanopore. Previously we have studied in detail the structural organization of $[EMIM][BF_4]$ in slit pores of various widths [34] and shown that the confined RTIL has a layering structure of less than 5 nm in width inside the pore. Therefore, in this section we will shortly consider the structure of other RTILs paying attention to the structural differences of the liquids associated with the alkyl chain length of the cation and the size and nature of the anion. General information about the adsorption of cations and anions can be obtained from the mass, number, and charge density profiles. Figs. 4-6 present the density profiles for the $[OMIM][NTf_2]$ ions. The profiles for $[EMIM][NTf_2]$ and $[OMIM][BF_4]$ are shown in Fig. 1S-6S in Supplementary

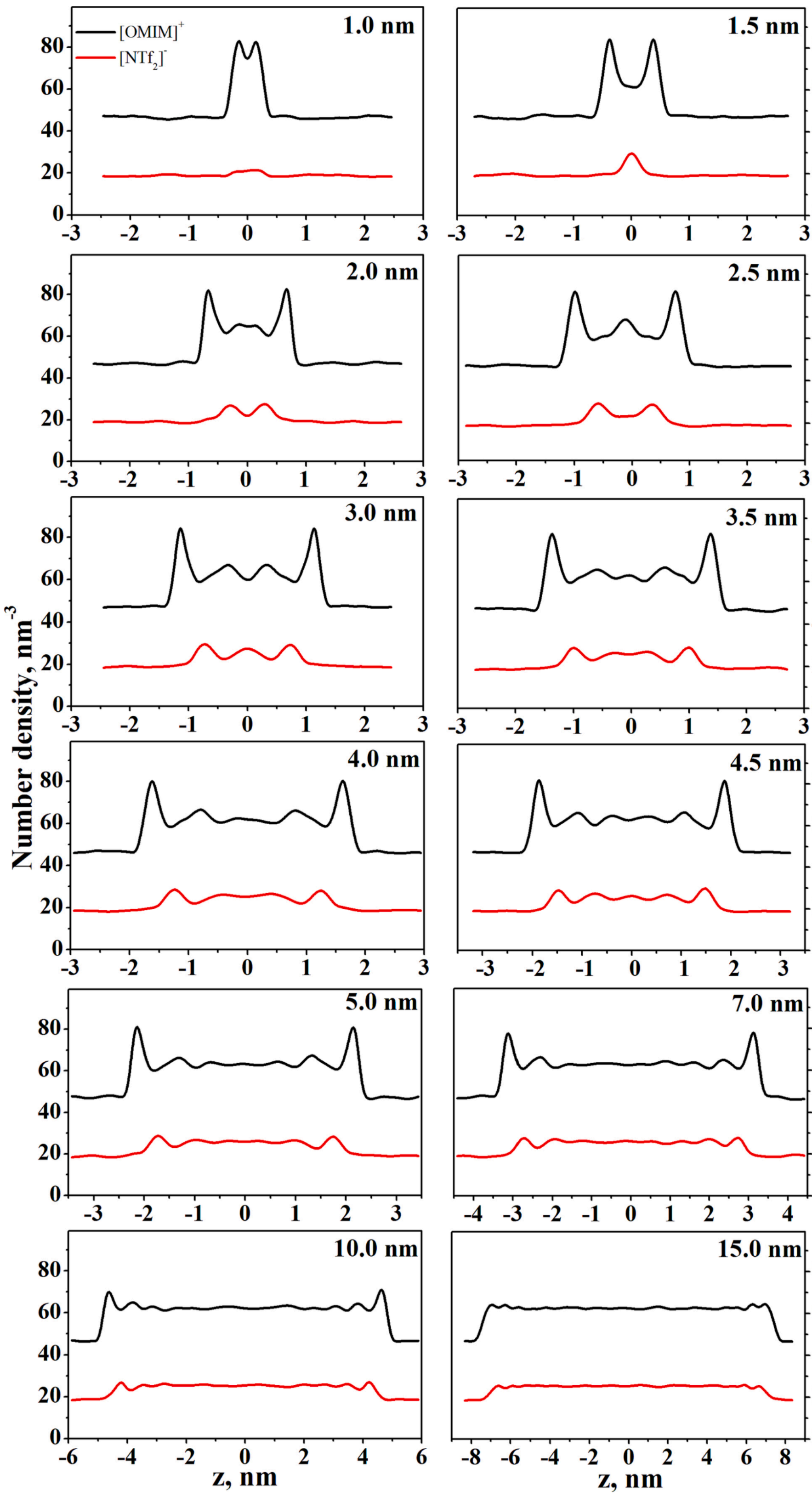


Fig. 5. Number density distribution profiles of [OMIM][NTf$_2$] in the direction perpendicular to the graphene surfaces at 300 K for h = 1–15 nm. The average is taken over the whole box.

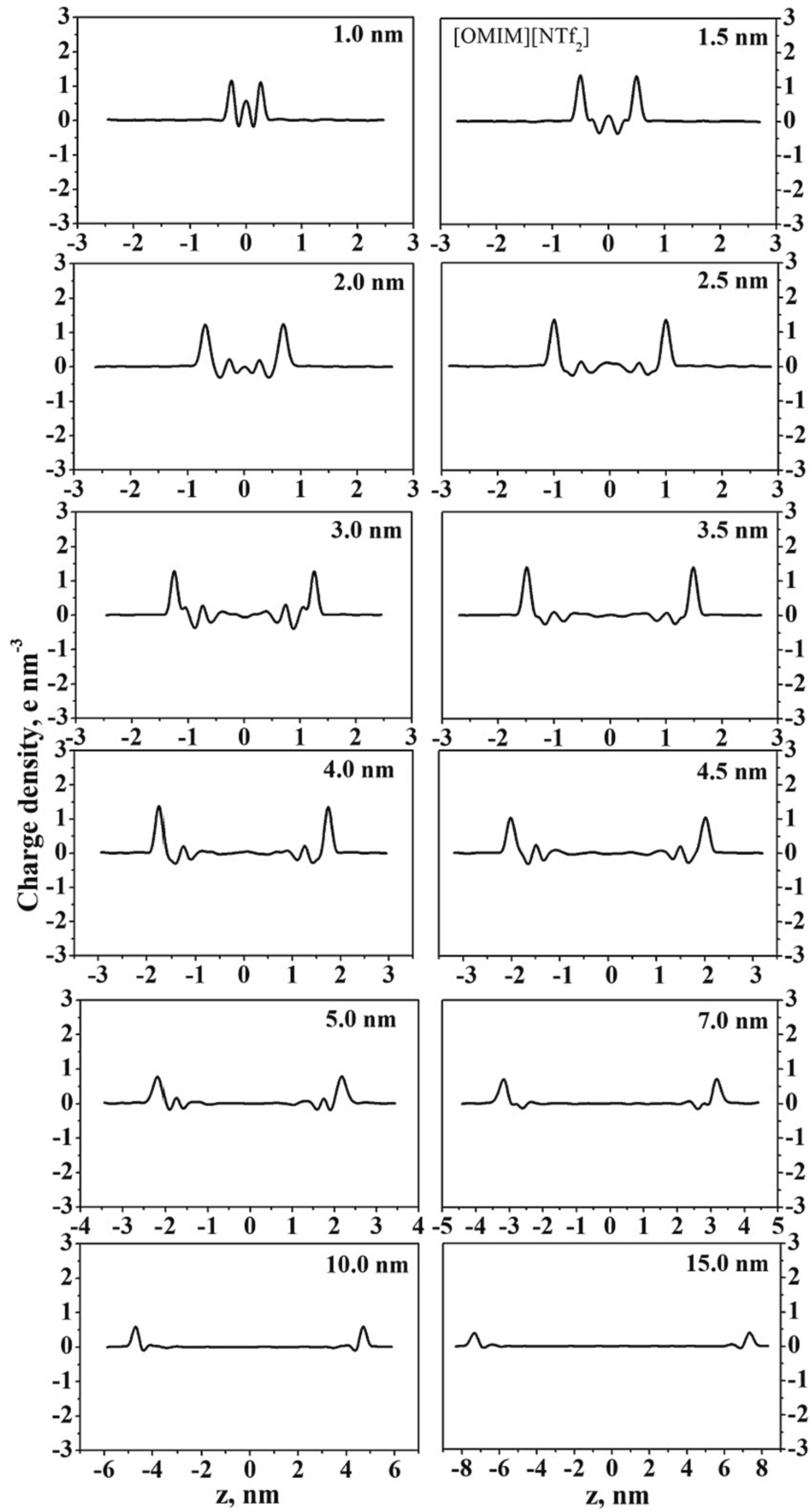


Fig. 6. Excess charge density distribution profiles of the [OMIM][NTf$_2$] cations and anions within negatively charged slit-shaped pores at the surface charge density of -0.8 e/nm^2 (-12.8 μC/cm^2) in the direction perpendicular to the graphene surfaces.

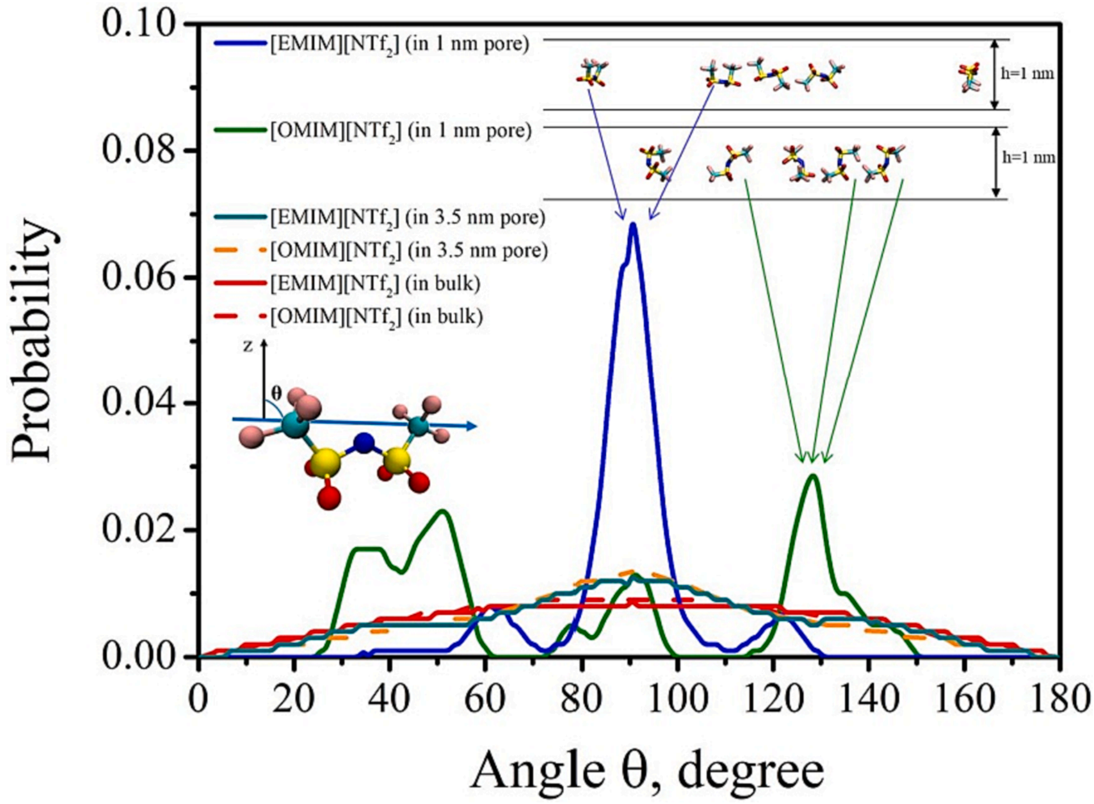


Fig. 7. Normalized probability distribution of the angle between the vector connecting the carbon atoms of $[NTf_2]^-$ and axis Z.

Material. Despite the fact that the average density inside the pore, ρ_p, is smaller than in the bulk, the RTIL mass density in the direction perpendicular to the graphene sheets demonstrates extreme behavior (Fig. 4). The maxima for all the RTILs are observed near the pore walls up to h = 10 nm. For the 15-nm-wide pores, these maxima almost disappear and the mass density distribution is uniform. As indicated in Fig. 5, $[OMIM]^+$ adsorbed onto graphene forms two prominent peaks, which are approximately 0.36 nm away from the innermost graphene sheets. These two well-defined peaks and the first peaks on the anion number density profiles for pores with $h \geq 2$ nm are considered to be the adsorption double-layer. For narrow pores with the width of 1 and 1.5 nm, we cannot distinguish the double electrical layer at each wall separately, since there is practically no free space for the anions as the cations are rather large in size. For $[EMIM]^+$-based RTILs the oscillation behavior of number density is observed up to 7 nm, and for $[OMIM]^+$-based RTILs the oscillation in the central part of the pore decays already at h = 5 nm. This is explained by the presence of a long movable alkyl chain in the $[OMIM]^+$ cation. To understand the adsorption, it is necessary not only to investigate the mass and number density profiles, which provide information about the overall position of the ions, but also to consider the charge density profiles, which can provide insight into the pore space from the electric neutrality point of view. Fig. 6 shows the excess charge density profiles of the ions inside the slit pore. The excess charge density indicates the presence of multiple charge layers along the Z direction. For $[NTf_2]^-$-based RTILs, complete compensation of positive and negative charges in the pore center is observed already at h = 5 nm. At the same time, for RTILs with a $[BF_4]^-$ anion noticeable excess charge oscillations are still registered in the pore center at h = 5 nm. Thus, based on the analysis of the mass, number and excess charge densities it can be argued that for all the considered RTILs there is a bulk region in the pore center, starting from the pore width of 7 nm.

In our previous work, we discussed the structure of $[EMIM][BF_4]$ inside the pore in detail and showed that the cations tended to have a parallel orientation on the charged graphene surface and the structure of the RTIL confined in a 1 nm pore not only differed from that in the bulk RTIL but was also different from the structure of the RTIL confined in wider pores. The current work also analyses the radial distribution functions (Fig. 7S-11S) and probability distribution of the angles between axis Z and different vectors (Fig. 12S-15S) in the ions of the RTILs under consideration. A detailed discussion can be found in the Supplementary Materials. Here we only draw the readers' attention to the differences in the structure of $[EMIM][NTf_2]$ and $[OMIM][NTf_2]$ located in a 1 nm pore. Fig. 7 presents the normalized probability distributions of the angle, θ, between the vector connecting the carbon atoms of the $[NTf_2]^-$ and the vector normal to the graphene surface as well as snapshots of instant orientation of the anions in 1 nm pores. As one can see, the distributions in the bulk phase and in the 3.5 nm wide pores demonstrate the same behavior for the two RTILs. But the distributions in the 1-nm-wide pore are quite different. For $[EMIM][NTf_2]$ there is a high peak at 90 degrees corresponding to the *cis*-conformer parallel to the graphene surfaces. The two symmetrical peaks at 60 and 120 degrees correspond to the *trans*-conformer of the anion. Due to the presence of a long alkyl chain and the need to compensate for the excess charge of the graphene surface the fraction of anions is small in the 1 nm pore filled with $[OMIM][NTf_2]$. The 1 nm pore filled with $[OMIM][NTf_2]$ contains a small fraction of anions due to the presence of a long alkyl chain and the need to compensate for the excess charge of the graphene surface. Since there is not enough free space and at the same time it is necessary to compensate for the strong electrostatic repulsion between the cations, the anions are predominantly in the *trans*-configuration.

3.2. Self-diffusion

The translational motions of cations and anions in the bulk phase, as well as in the pore, can be investigated through the mean squared displacement (MSD). The self-diffusion coefficients (D) can be computed employing Einstein's relation:

Table 1
Self-diffusion coefficients of the cations (D_C) and anions (D_A), summation of the cationic and anionic self-diffusion coefficients (D_{C+A}), transference numbers of the cations and anions bulk phase of [EMIM][BF$_4$], [OMIM][BF$_4$], [EMIM][NTf$_2$] and [OMIM][NTf$_2$] according to MD and literature data.

RTIL		Self-diffusion coefficient, 10^{-11} m^2/s			Transference number	
		MD data	Literature data		MD data	Literature data
[EMIM][BF$_4$]	D_C	0.72 ± 0.12	1.10[77], 5.32[78]	t_C	0.66	0.55[77], 0.54[78]
	D_A	0.37 ± 0.03	0.90[77], 4.46[78]	t_A	0.34	0.45[77], 0.46[78]
	D_{C+A}	1.09 ± 0.08	2.00[77], 9.78[78]			
[OMIM][BF$_4$]	D_C	0.64 ± 0.04	0.80[77]	t_C	0.65	0.57[77]
	D_A	0.35 ± 0.02	0.60[77]	t_A	0.35	0.43[77]
	D_{C+A}	0.99 ± 0.03	1.40[77]			
[EMIM][NTf$_2$]	D_C	4.63 ± 0.18	5.31[36]	t_C	0.58	0.62[36]
	D_A	3.38 ± 0.22	3.32[36]	t_A	0.42	0.38[36]
	D_{C+A}	8.01 ± 0.20	8.63[36]			
[OMIM][NTf$_2$]	D_C	1.85 ± 0.13	1.29[36]	t_C	0.53	0.50[36]
	D_A	1.65 ± 0.01	1.28[36]	t_A	0.47	0.50[36]
	D_{C+A}	3.50 ± 0.07	2.57[36]			

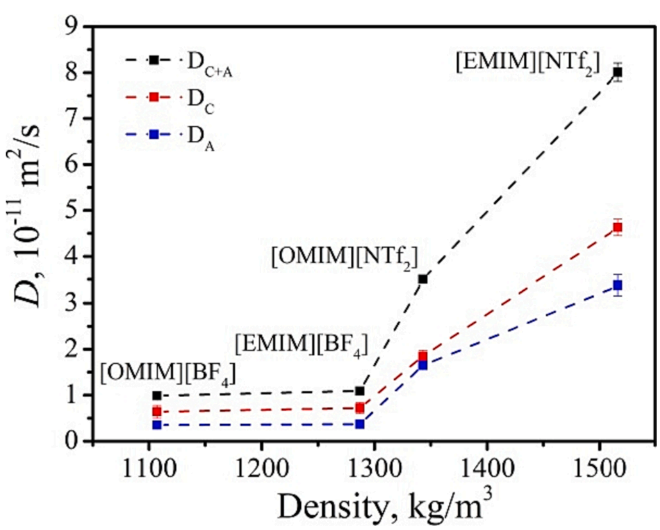


Fig. 8. Dependence of self-diffusion coefficients of the cations, anions and summation of the cationic and anionic self-diffusion coefficients for [EMIM][BF$_4$], [OMIM][BF$_4$], [EMIM][NTf$_2$] and [OMIM][NTf$_2$] on the density of the bulk RTIL at 300 K.

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{\left\langle \left|r_i(t) - r_i(0)\right|^2 \right\rangle}{t} \quad (1)$$

where $\left\langle \left|r_i(t) - r_i(0)\right|^2 \right\rangle$ is the MSD in the XYZ space; in brackets are the values averaged over time and number of cations and anions. Previously such approach was used for MD studies of the temperature dependence of self-diffusion of bulk RTILs [57] and RTILs in contact with an air surface [58], as well as for investigation of self-diffusion of different species including RTILs in slit and cylindrical pores [17,23,59–63]. It is important to mention that the Einstein relation we utilized is solely applicable for calculating self-diffusion coefficients in bulk ILs. This limitation arises from the fact that certain assumptions made by Einstein's relation, such as the homogeneous and isotropic nature of the medium, may not hold true for confined ILs. The spatial distribution of particles in equilibrium exhibits nonuniformity, which renders methods based on the conventional diffusion equation neglecting density spatial variations are inadequate. Numerous studies have proposed and utilized different approaches to determine diffusivity profiles in inhomogeneous systems and confined fluids [64–71]. In this study, we investigate the diffusion of confined ILs by analyzing their mean square displacement. Firstly, we calculated the self-diffusion coefficients of the RTIL ions in the bulk phase to make sure that the chosen force field parameters adequately reproduce the dynamic properties. Our results predict that the diffusion coefficients of the ions in ionic liquids are of the order of 10^{-11} m^2/s, which is in agreement with the results of the other computational and experimental studies summarized in Table 1. Fig. 8 clearly indicates that the summed values of the cationic and anionic self-diffusion coefficients tend to increase as follows: D[OMIM][BF$_4$] < D[EMIM][BF$_4$] < D[OMIM][NTf$_2$] < D[EMIM][NTf$_2$]. Notably, an increase in the RTIL density leads to higher values of the self-diffusion coefficients. Also the magnitude of the self-diffusion coefficients is shown to be strongly dependent on the length of the alkyl group on the imidazolium cation and on the anion nature. As expected, the smaller [EMIM]$^+$ cation diffuses more quickly than the large [OMIM]$^+$ cation with its long alkyl chain which limits its mobility. This result is in good agreement with the previous experimental observations on the imidazolium-based ionic liquids [36,72–76]. The opposite effect is observed for anions. The smaller [BF$_4$]$^-$ anion demonstrates slower diffusion than the [NTf$_2$]$^-$ anion, which can be associated with higher surface electrical charge density than that of [NTf$_2$]$^-$ (Fig. 9) and, as a result, higher energy of interaction with the cation (Fig. 10 and Table 6S) and good packing efficiency in the RTIL. The interaction energy calculated for the [EMIM]$^+$ and [OMIM]$^+$ complexes with [NTf$_2$]$^-$ is substantially smaller than that for the complexes with [BF$_4$]$^-$, which suggests that the self-diffusion coefficients depend on the magnitude of the anion-cation interaction. Besides, the [OMIM]$^+$ and [EMIM]$^+$ cations have higher self-diffusion coefficients than those of the anions, even if the cationic radius and molar mass are bigger than those of the anion (in case of [EMIM][BF$_4$] and [OMIM][BF$_4$]), which makes the imidazolium cations better electron carriers than the anions despite their molecular size. In literature this phenomenon was attributed to the planar structure of the imidazolium cations and the greater friction of the translational motions of the anions than that of the planar imidazolium cations [76].

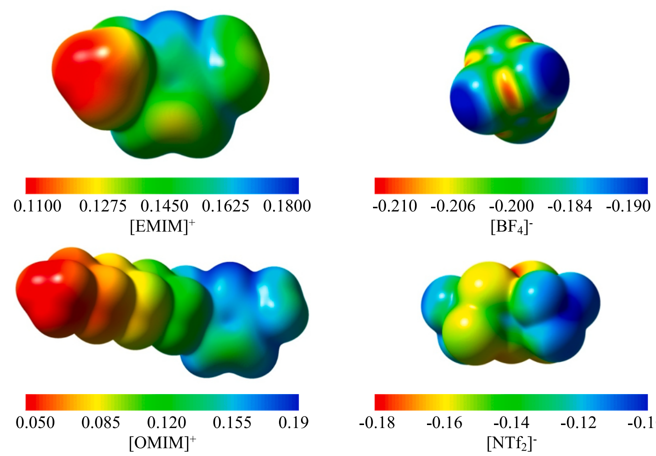
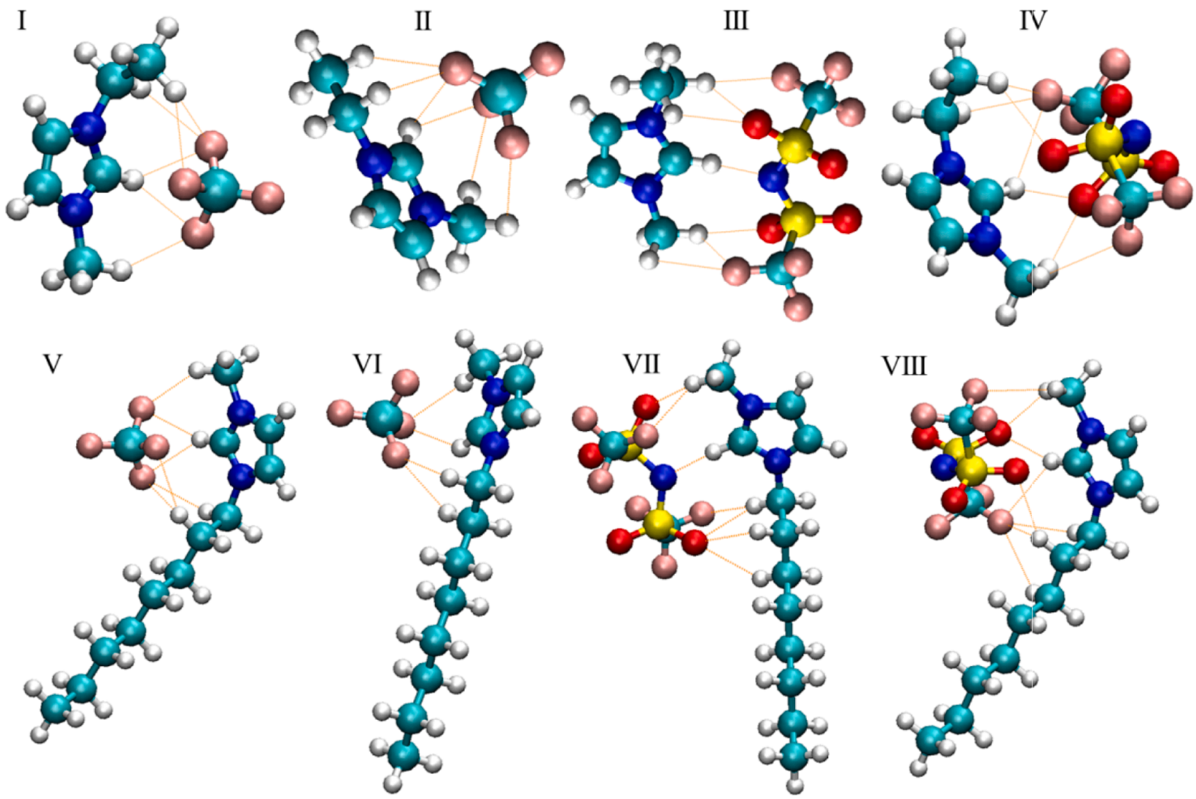


Fig. 9. Cation and anion ESP-mapped total density (0.001) surfaces. Detailed discussion is presented in Supplementary Material.

The relative contribution of the charged species to the total charge



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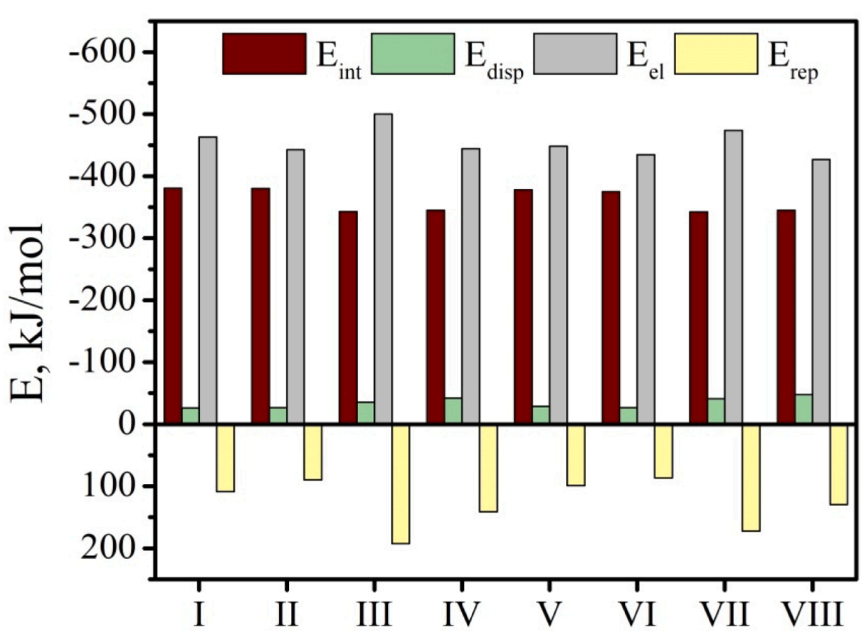

b)

Fig. 10. Optimized geometries for [EMIM][BF_4] (I-II), [EMIM][NTf_2] (III-IV), [OMIM][BF_4] (V-VI), [OMIM][NTf_2] (VII-VIII) (a); interaction energy (E_{int}), dispersion (E_{disp}), electrostatic (E_{el}) and repulsion (E_{rep}) energies for structures I-VIII (b). Detailed discussion is presented in Supplementary Material.

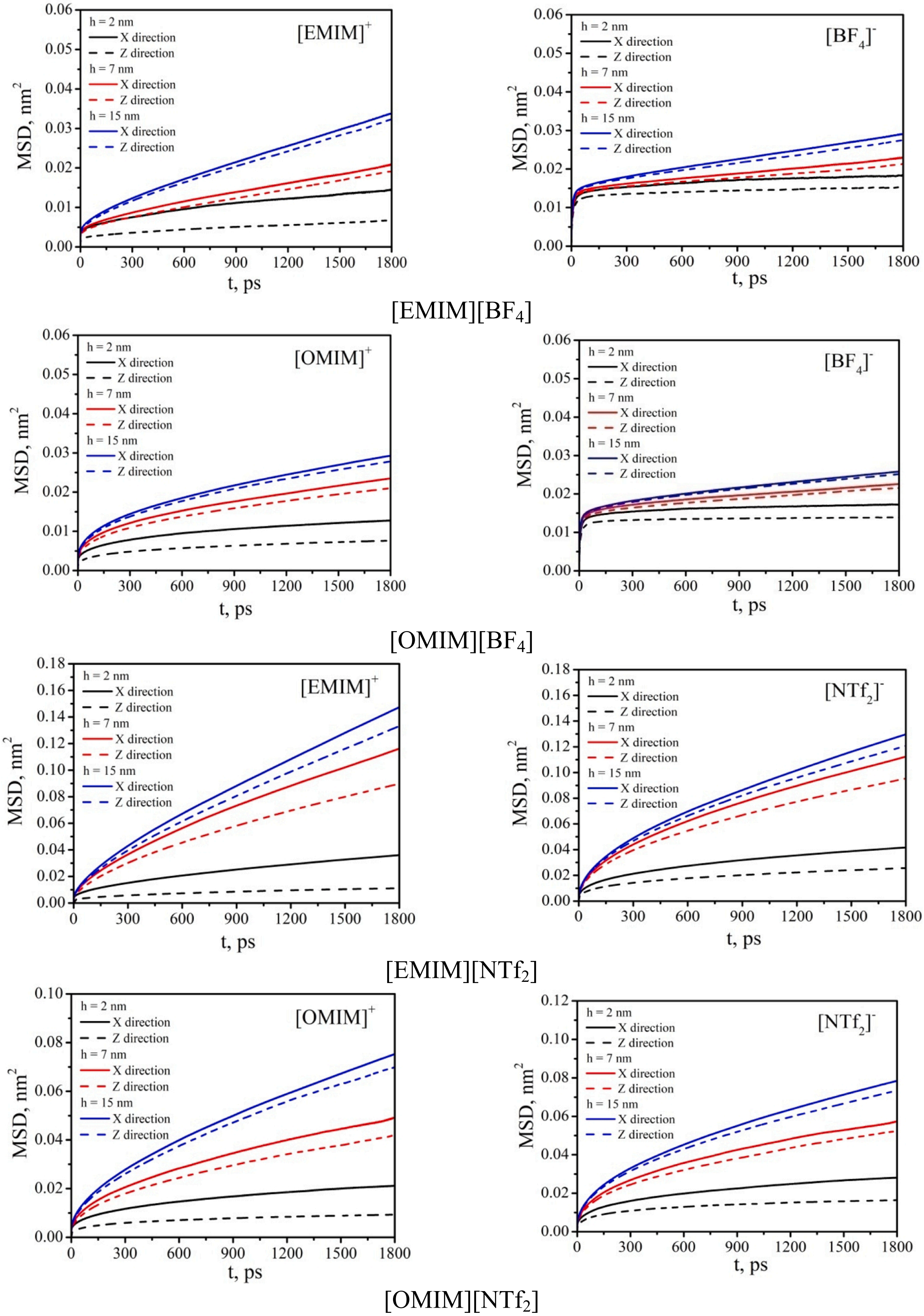

Fig. 11. Parallel (x) and normal (z) components of the mean squared displacements (MSDs) of cations and anions of RTILs confined inside 2-, 7-, and 15-nm-wide slit carbon nanopores.

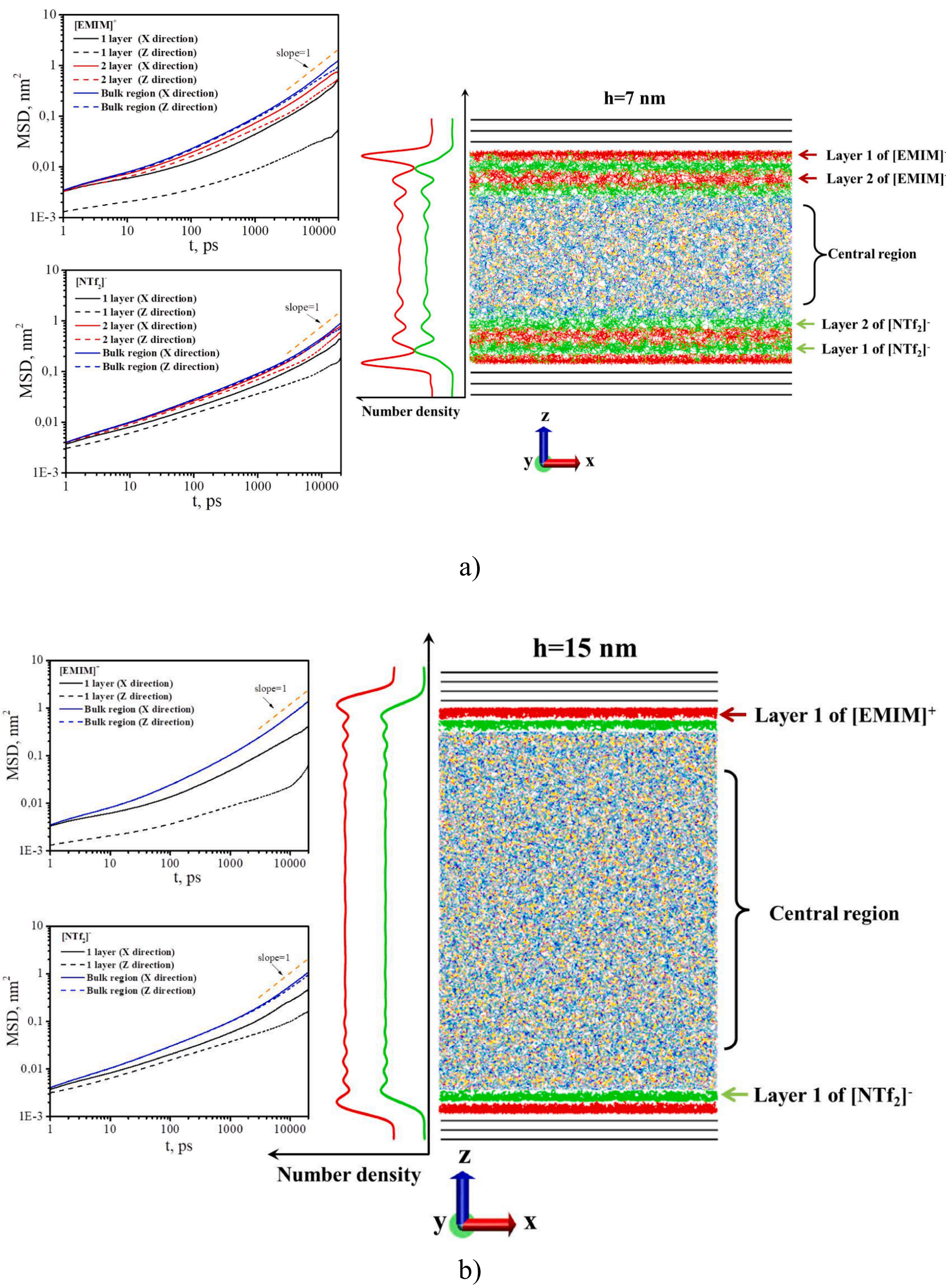

Fig. 12. Final configuration of [EMIM][NTf_2] confined in 7 nm (a) and 15 nm (b) pores. In the center are the illustrative equilibrium density profiles of cations (red) and anions (green) along the Z direction. On the left is the log–log plot of the parallel and normal components of the MSD of the confined cations and anions in different regions inside the pore. The dashed orange line indicates the behavior expected in the Fickian regime (MSD ∝ t).

transfer is one of the important properties of ionic liquids, which are used as electrolytes in electrochemical processes and devices. For simple 1:1 electrolytes consisting of monovalent cation and anion, the transference numbers can be found from the self-diffusion coefficients of the cation and anion:

$$t_C = \frac{D_C}{D_{C+A}}, t_A = \frac{D_A}{D_{C+A}} \tag{2}$$

where D_C and D_A are the self-diffusion coefficients of the cations and anions, and D_{C+A} is the summation of the cationic and anionic self-diffusion coefficients. The calculated cationic transference numbers

Table 2
Mean squared displacement at 10 ns for the RTIL cations and anions in different regions inside 7-nm- and 15-nm-wide pores in the X (MSD_x), Y (MSD_y), and Z (MSD_z) directions.

Region	Cation			Anion		
	MSD_x, nm^2	MSD_y, nm^2	MSD_z, nm^2	MSD_x, nm^2	MSD_y, nm^2	MSD_z, nm^2
	[EMIM][BF_4]					
	h = 7 nm					
Layer 1	0.02	0.02	0.005	0.02	0.02	0.01
Layer 2	0.04	0.04	0.01	0.03	0.03	0.03
Central	0.05	0.05	0.05	0.03	0.03	0.03
	h = 15 nm					
Layer 1	0.02	0.02	0.005	0.02	0.02	0.01
Central	0.07	0.07	0.07	0.05	0.05	0.05
	[OMIM][BF_4]					
	h = 7 nm					
Layer 1	0.03	0.03	0.01	0.03	0.03	0.02
Layer 2	0.04	0.05	0.02	0.04	0.04	0.03
Central	0.07	0.07	0.07	0.05	0.05	0.05
	h = 15 nm					
Layer 1	0.02	0.03	0.01	0.02	0.03	0.01
Central	0.08	0.10	0.08	0.06	0.07	0.06
	[EMIM][NTf_2]					
	h = 7 nm					
Layer 1	0.24	0.26	0.03	0.23	0.25	0.11
Layer 2	0.43	0.45	0.30	0.42	0.39	0.28
Central	0.63	0.70	0.54	0.46	0.48	0.42
	h = 15 nm					
Layer 1	0.24	0.26	0.02	0.28	0.30	0.10
Central	0.70	0.72	0.70	0.56	0.54	0.50
	[OMIM][NTf_2]					
	h = 7 nm					
Layer 1	0.07	0.07	0.02	0.09	0.11	0.05
Central	0.20	0.23	0.20	0.16	0.20	0.19
	h = 15 nm					
Layer 1	0.07	0.08	0.02	0.10	0.11	0.05
Central	0.26	0.28	0.27	0.24	0.23	0.23

(Table 1) are higher than 0.5 in all the cases, which is also consistent with the literature data.

The presence of charged graphene surfaces complicates the RTILL dynamic behavior caused by preferred orientations and reduced mobility. To gain insight into the dynamics of the RTILs inside the pore, Fig. 11 presents the MSD of the RTIL ions in the X (parallel to the surface) and Z (normal to the surface) directions for the pore sizes of 2, 7, and 15 nm. Unlike the bulk RTILs, the effects of confinement cause inherent inhomogeneity in confined RTILs due to the interactions between the ions and the charged surfaces. The inhomogeneity, however, gradually disappears with an increase in the pore width and appearance of a layer close to the bulk phase. For the cations and anions of all the RTILs one can observe (Fig. 11) a decrease in the discrepancy of the MSD value in the X and Z directions as the pore becomes wider. Notably, for cations, especially in small pores, the discrepancy in the MSD_X and MSD_Z values is bigger than for anions due to the stronger interactions of the former with the graphene surface. Comparing the MSD for 2 nm and 7 nm pores one can note that the cations move faster than the anions (as in the bulk systems) only in the large pores, where the fraction of cations in the bulk-like central part significantly exceeds the number of cations near the pore walls. The mobility of cations and anions increases in the series of RTILs in the following order: [OMIM][BF_4] < [EMIM][BF_4] < [OMIM][NTf_2] < [EMIM][NTf_2]. Besides, one can see that the confinement significant hinders the diffusion of the RTILs. This is to be expected since as h goes to infinity, the effect of confinement must become negligible.

Studying [EMIM][NTf_2] as an example, we demonstrate that the dynamics of a confined RTIL is heterogeneous. In Fig. 12 we show a log–log plot of the parallel (in the X direction) and normal (in the Z direction) components of the MSD of the confined cations and anions in the first and/or second layers closest to the walls and in the central

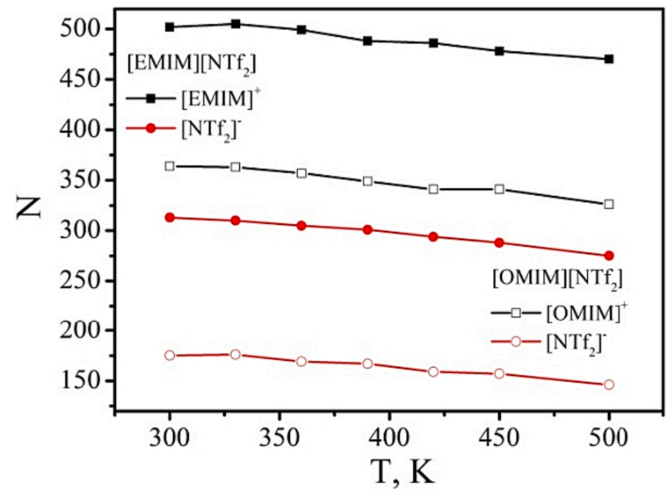

Fig. 13. Temperature dependence of the ion number in a 2-nm-wide pore for [EMIM][NTf_2] and [OMIM][NTf_2].

region where the RTIL structure is close to the bulk phase. The slowest dynamics are observed for the ions in the first layers, and the dynamics of the ions in the pore center are very similar to those observed for the ions in the bulk. These results are in agreement with those obtained for [BMIM][PF_6] in the multiwalled carbon nanotubes [23] and for supercooled LJ fluids in confinement [79]. The visual analysis of the trajectories shows that the cations of the first layer do not move in the Z direction and are only seen to oscillate during the simulation procedure. The MSD at 10 ns for the cations and anions in various regions within pores of different widths (7-nm and 15-nm) is provided in Table 2. It is observed that, in the central part of the pore, the X, Y, and Z components of the MSD increase as the width (h) increases. Moreover, these components become almost equal to each other, suggesting that the structure of the ILs in the central part of the pore becomes similar to that of the bulk ILs.

Studying [EMIM][NTf_2] and [OMIM][NTf_2] confined in a 2-nm-wide pore as an example, we attempted to determine the effect of temperature on the RTIL dynamics in the wide range from 300 K to 500 K. The dependence of the self-diffusion coefficient of all the RTILs in the bulk was also calculated. The RTIL density is known to decrease as the temperature goes up and, as a consequence, the number of ions inside the pore also decreases (Fig. 13). At the same time, the pore loading increases from 78.4 % to 80.5 % and from 75.6 % to 77.6% of the RTIL bulk density for [EMIM][NTf_2] and [OMIM][NTf_2], respectively. Figs. 14 and 15 show the MSD of the ions inside the pore in the XY and Z directions and in the bulk versus time in logarithmic-logarithmic representation in the temperature range of 300–500 K. Two regimes are observed in the MSDs reported in Figs. 14 and 15: the sub-diffusive regime [80], where the ions are trapped within a cage formed by their neighbors (MSD $\propto t^{\alpha}$, with $0 < \alpha$ less than 1), and the diffusive regime, in which the diffusion is characterized by MSD $\propto$ t, when the ions escape from these cages. The longest sub-diffusive regime is observed for the z-component of MSD and the diffusive regime is only achieved at the temperature of 390 K for $[EMIM]^+$, 450 K for $[OMIM]^+$, and 420 K for $[NTf_2]^-$. Long sub-diffusive regime in this case, when the ions do not exhibit the typical random, Brownian motion and instead exhibit hindered diffusion due to the presence of strong electrostatic interactions between the ions and the confining surface, orientational preference of cations in interfacial regions and as a consequence the formation of structured and highly ordered environment within the confinement (3 layers of cations and 2 layers of anions [34]) that restrict the motion of the ions. Previously sub-diffusive behaviors have been reported in the literature for CNTs filled with ILs [81], for ILs in confined films [82] and in slit nanopore [17], for gas inside nanoporous carbon structures

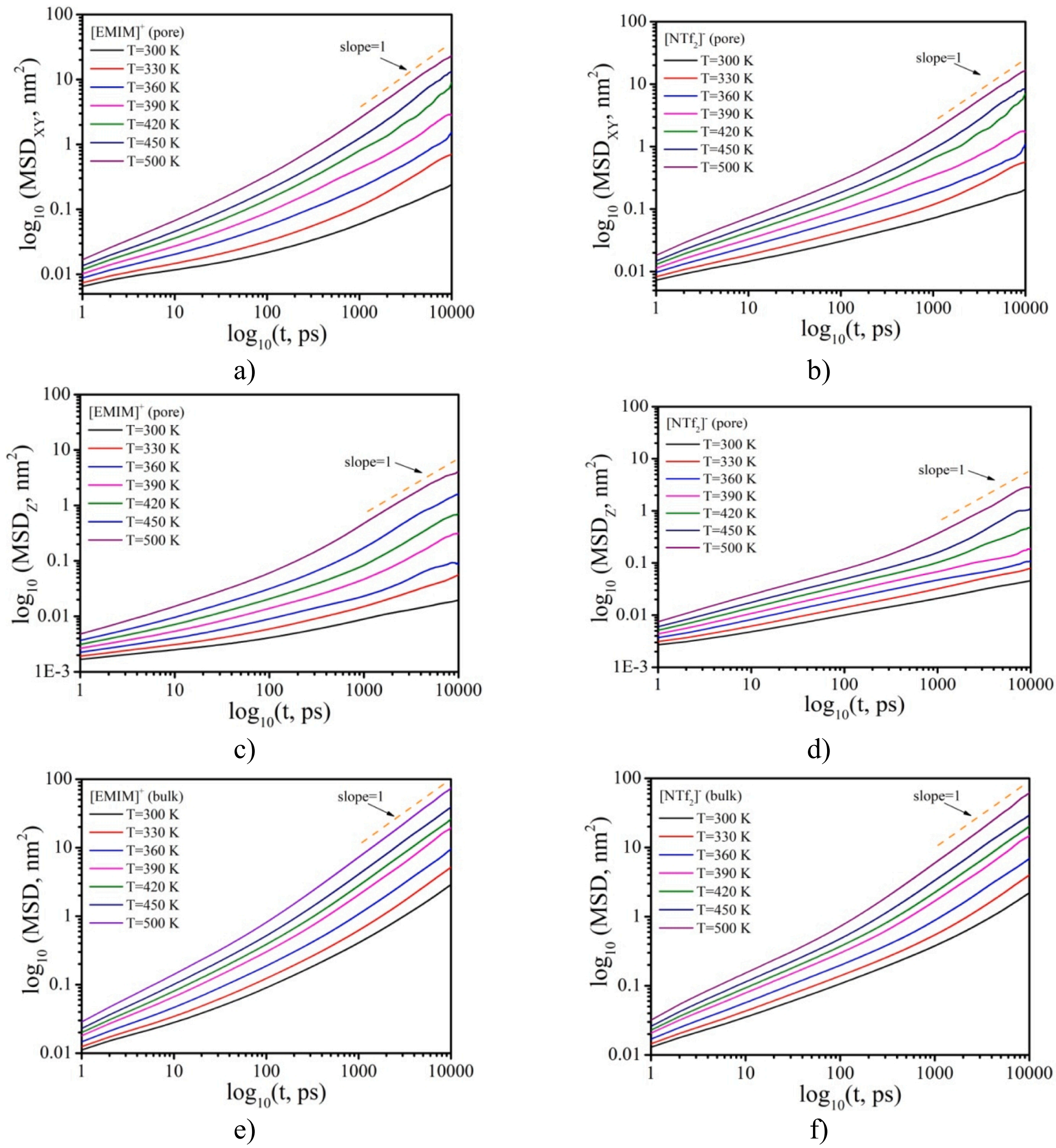

Fig. 14. Mean squared displacements of $[EMIM]^+$ and $[NTf_2]^-$ ions in a 2-nm-wide pore in the XY and Z directions and in the bulk versus time in logarithmic-logarithmic representation in the temperature range of 300–500 K.

[83,84], for binary Lennard-Jones liquid confined between two parallel rough walls [79]. For the XY-component of MSD and for MSD of the ions in the bulk, the diffusive regime is achieved faster and at a lower temperature. Despite the fact that the pore loading insignificantly increases with temperature, the RTIL structure becomes more disordered. Although the layered structure is still preserved at high temperatures (Fig. 16a), it becomes less pronounced compared with those at 300 K (Fig. 2). The peaks in the number and charge density profiles become much lower (Fig. 16b and 16c), and the cations become capable not only of oscillating near the walls of the pore in the Z direction (which occurs at 300 K), but also of moving from one charged surface to the other (Fig. 16a). MSD at 10 ns for the cations and anions was used as the indication of the self-dynamics of the confined RTILs (Table 3). In case of $[EMIM][NTf_2]$ inside the pore, the cations move faster than the anions, as in the bulk, even despite the fact that the largest fraction of $[EMIM]^+$ is located near the charged surfaces (which tend to slow down the $[EMIM]^+$) and only one layer is in the central part of the pore. For the RTIL with a longer alkyl chain the opposite behavior is observed. As it was previously shown (Table 1), the self-diffusion coefficients of $[OMIM]^+$ and $[NTf_2]^-$ are almost equal to each other in the bulk phase. But inside the 2 nm pore the $[OMIM]^+$ dynamics are slower than those of the anions. And with temperature growth, the difference between the cation and anion mobility increases.

The summed anionic self-diffusion coefficients for all the RTILs in the bulk phase as a function of temperature is presented in Fig. 17. The trend of increasing values of the summed cationic and anionic self-diffusion coefficients discussed above for the systems at 300 K ($D[OMIM][BF_4] < D[EMIM][BF_4] < D[OMIM][NTf_2] < D[EMIM][NTf_2]$) holds at high

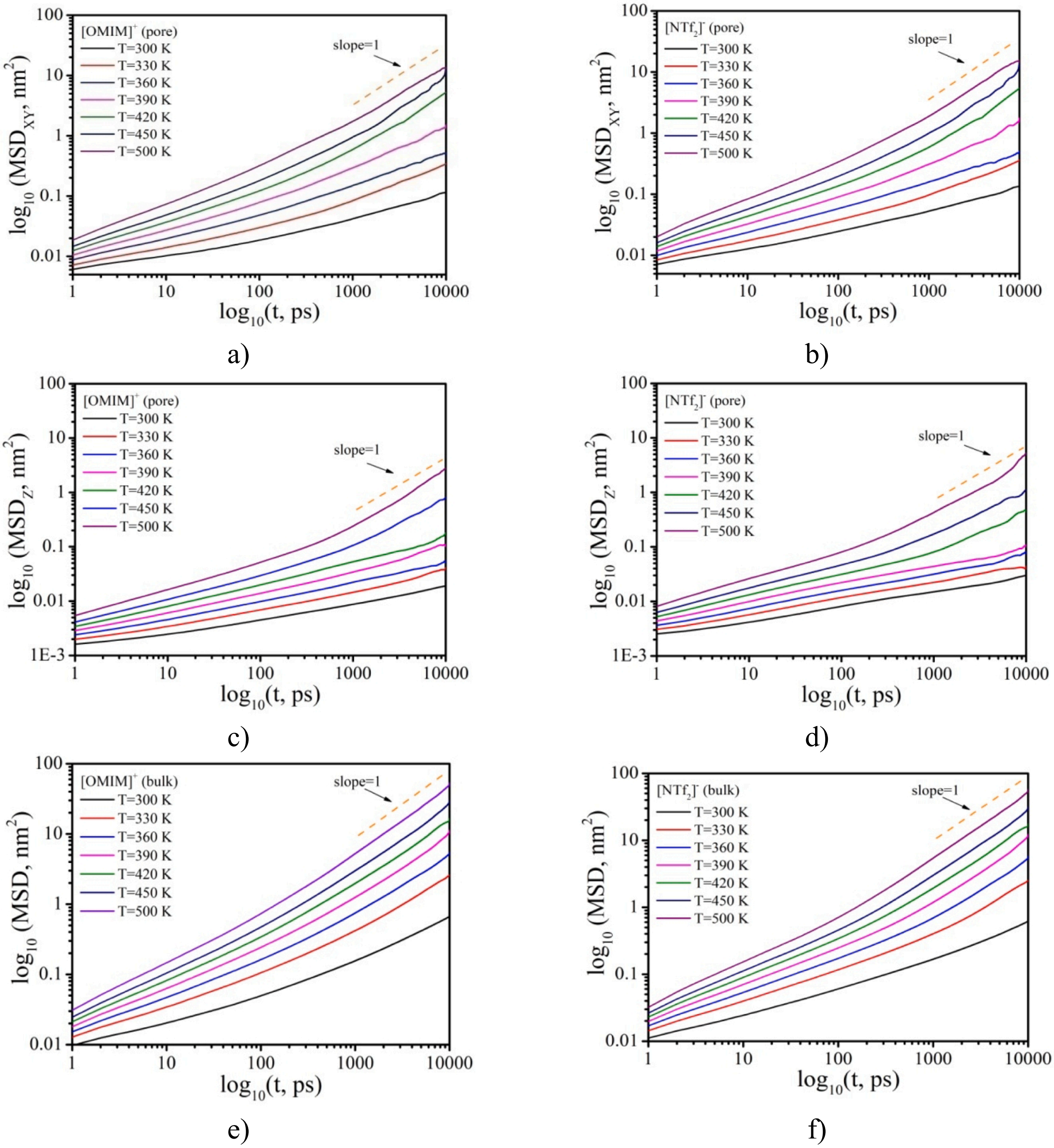

Fig. 15. Mean squared displacements of $[OMIM]^+$ and $[NTf_2]^-$ ions in a 2-nm-wide pore in the XY and Z directions and in the bulk versus time in logarithmic-logarithmic representation in the temperature range of 300–500 K.

temperatures. The temperature dependences of D_C, D_A, and D_{C+A} were fitted to the Arrhenius (ARH) equation $D = D_0\exp(-E_a/RT)$ and the Vogel-Fulcher-Tamman (VFT) equation $D = D_0\exp(-B/(T-T_0))$ for diffusivity, where E_a is the activation energy (kJ/mol), the constants D_0 (cm^2/s), B (K), and T_0 (K) are adjustable parameters (Fig. 18). As can be seen in Fig. 18 for $[EMIM][NTf_2]$ and $[OMIM][NTf_2]$, the MD data for the self-diffusion coefficients are in excellent agreement with the Arrhenius and Vogel-Fulcher-Tamman equations over the whole temperature range. For the RTILs with a $[BF_4]^-$ anion, the VFT equation gives a better approximation of D(T) than ARH. The activation energy in the ARH equation (i.e. the energy barrier which the ions have to overcome to travel by each other) allows analyzing the dependence of self-diffusion coefficients on the cation alkyl chain length and on the anion type. Analyzing the activation energy parameter in the ARH equation (Table 4) we can conclude that the higher value of E_a, the smaller the self-diffusion coefficient of the cations and anions and vice versa: the lower the Ea, the more easily the ions can move. For example, for $[BF_4]^-$ – based RTILs, E_a is considerably higher than that for $[NTf_2]^-$ – based RTILs resulting in slower dynamics of $[EMIM][BF_4]$ and $[OMIM][BF_4]$. This result is in agreement with the data of ionic pairs interaction energies obtained by quantum chemical calculations (Fig. 10 and Table 6S).

3.3. Electrical conductivity

As well as for calculating self-diffusion coefficients of ILs, there are

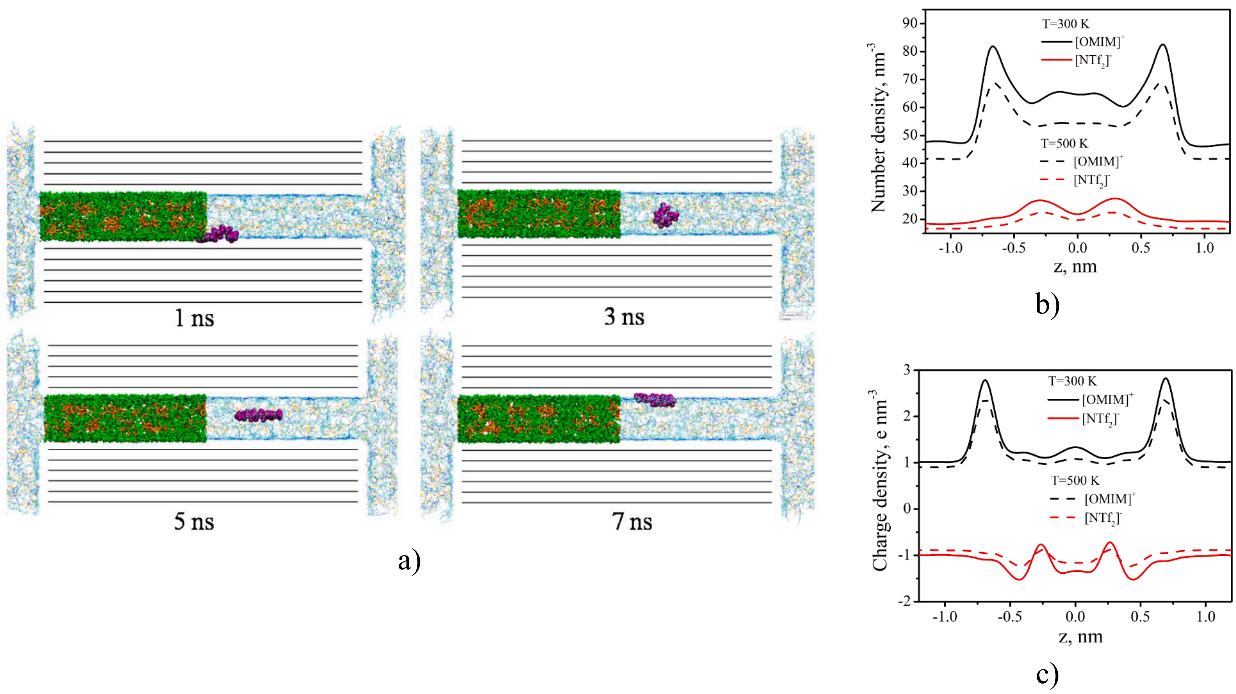

Fig. 16. (a) Instant snapshots of [OMIM][NTf$_2$] inside a 2-nm-wide pore at 500 K. In the pore region, the green and orange colors represent the [OMIM]$^+$ cations and [NTf$_2$]$^-$ anions. Black represents the carbon atoms of the graphene sheets. Purple represents the position of a random [OMIM]$^+$ cation during the simulation. The number (b) and the charge (c) density profiles of [OMIM][NTf$_2$] in the Z direction at 300 K and 500 K for h = 2 nm.

Table 3
Mean squared displacement at 10 ns for [EMIM][NTf$_2$] and [OMIM][NTf$_2$] ions confined inside a 2-nm-wide pore at 300 K – 500 K.

	[EMIM][NTf$_2$]		[OMIM][NTf$_2$]	
T, K	MSD (Cation), nm^2	MSD (Anion), nm^2	MSD (Cation), nm^2	MSD (Anion), nm^2
300	0.26	0.25	0.13	0.16
330	0.70	0.57	0.56	0.54
360	1.66	1.14	0.56	0.56
390	1.88	1.36	1.54	1.77
420	10.16	8.30	4.02	4.75
450	17.58	12.88	12.73	13.89
500	31.77	28.43	15.73	19.68

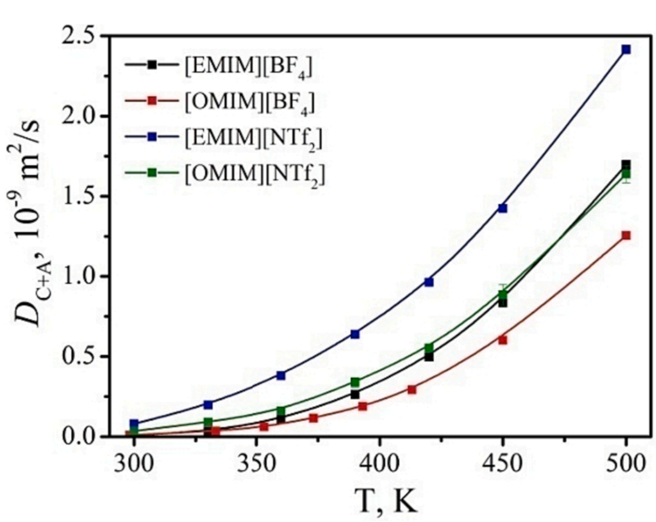

Fig. 17. Temperature dependence of the summed cationic and anionic self-diffusion coefficients of the RTIL ions in the bulk phase.

several methods for calculating their ionic conductivity. These methods include the Green–Kubo formalism [21], the Einstein–Helfand relation [85], the Stokes–Einstein relation [86], and the Nernst–Einstein relation [87]. The latter approach, where σ_{NE} is considered to be proportional to the weighted sum of separate ionic diffusion coefficients [88,89], is commonly used to quick estimate electrical conductivity:

$$\sigma_{NE} = \frac{e^2}{kT}\sum_{i=1}^{N}\rho_i q_i^2 D_i \tag{3}$$

where e is the electric charge unit, 1.60219 * 10^{-19} C, and ρ_i is the density of species i. Previously, studies by Lee et al. [87] calculated the ionic conductivity of RTILs using the Nernst-Einstein relation and found that the calculated values accurately reproduced the experimental trend. They concluded that the self-diffusion coefficient of ionic liquids is the dominant factor to determine the ionic conductivity of ionic liquids. However, it is important to note that the Nernst-Einstein ionic conductivity does not account for the contribution from the cross term (D_CD_A), and therefore, it is not the true ionic conductivity. The work [90] provides the relationship between the ionic conductivity calculated using the Green-Kubo approach and the Nernst-Einstein ionic conductivity. This relationship incorporates the deviation parameter, Δ, which is defined based on the time integral of the velocity cross correlation function for unlike ions. It is independent of temperature and pressure, ionic masses, and experimentally, it appears to be influenced by the size and shape of the ions [91].

As it was discussed above, increasing the alkyl chain length limits the movement of the imidazolium cations, which means the RTILs with a longer alkyl chain [OMIM][BF$_4$] and [OMIM][NTf$_2$] have lower electrical conductivities than [EMIM][BF$_4$] and [EMIM][NTf$_2$], respectively. As expected, [EMIM][NTf$_2$], whose ions have higher self-diffusion coefficients, has the highest electrical conductivity of all the RTILs under consideration in this study. At the same time, despite having lower ionic diffusion coefficient values than [OMIM][NTf$_2$], [EMIM][BF$_4$] demonstrates higher electrical conductivity. This is due to its high concentration of charge carriers (Fig. 3a). The experimental σ

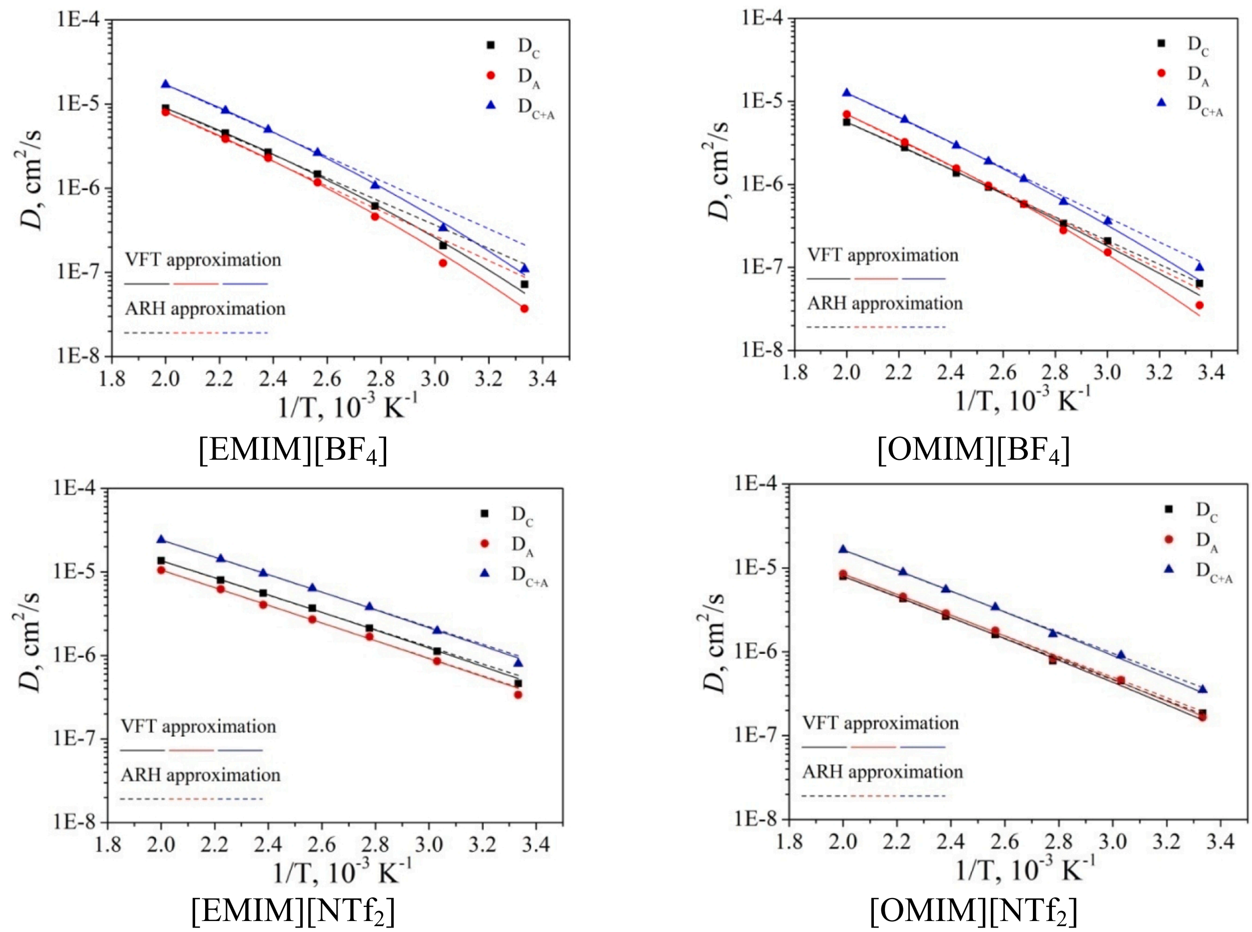

Fig. 18. Temperature dependence of the cationic, anionic and summed self-diffusion coefficients with ARH and VFT approximation.

Table 4
ARH and VFT equation parameters of the self-diffusion coefficient data.

RTIL		ARH		VFT		
		D_0, 10^{-4} cm^2/s	E_a, kJ/mol	D_0, 10^{-4} cm^2/s	B, 10^2 K	T_0, K
[EMIM][BF$_4$]	C	54.1 ± 7.1	26.6 ± 0.5	7.4 ± 1.6	16.5 ± 1.4	125 ± 14
	A	71.0 ± 9.7	28.2 ± 0.5	8.9 ± 3.2	17.6 ± 2.4	125 ± 22
	C + A	122.4 ± 16.0	27.3 ± 0.5	16.1 ± 3.8	17.1 ± 1.5	125 ± 14
[OMIM][BF$_4$]	C	40.2 ± 3.2	27.3 ± 0.3	12.8 ± 5.9	23.4 ± 3.5	69 ± 28
	A	89.9 ± 8.8	29.8 ± 0.4	13.1 ± 2.0	20.5 ± 1.1	109 ± 9
	C + A	123.8 ± 10.1	28.6 ± 0.3	26.4 ± 6.5	21.9 ± 1.8	90 ± 14
[EMIM][NTf$_2$]	C	15.6 ± 0.77	19.7 ± 0.2	10.4 ± 3.2	20.2 ± 2.5	33 ± 25
	A	13.6 ± 1.0	20.2 ± 0.3	12.0 ± 7.4	23.2 ± 5.3	10 ± 48
	C + A	29.1 ± 1.2	19.9 ± 0.2	21.9 ± 6.4	21.5 ± 2.4	23 ± 23
[OMIM][NTf$_2$]	C	23.4 ± 1.8	23.6 ± 0.3	12.6 ± 6.9	23.2 ± 4.5	43 ± 38
	A	24.5 ± 1.7	23.5 ± 0.3	14.6 ± 7.6	23.9 ± 4.8	36 ± 36
	C + A	47.8 ± 3.2	23.6 ± 0.3	27.2 ± 13.0	23.5 ± 3.9	39 ± 33

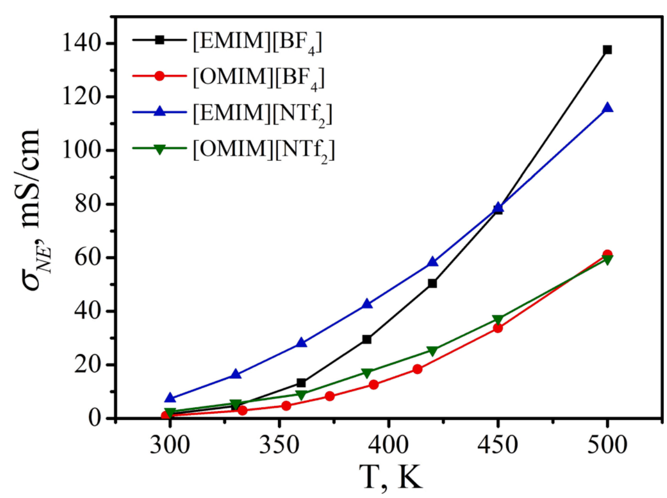

Fig. 19. Electrical conductivity of RTILs in the bulk from the Nernst–Einstein relation as a function of temperature.

values obtained for [EMIM][NTf$_2$] and [OMIM][NTf$_2$] are 9.34 mS/cm and 1.44 mS/cm [36], respectively, which is in agreement with the MD data. Comparing σ for RTILs with different anions one can note that the RTIL with a larger [NTf$_2$]$^-$ anion has higher conductivity than that with a smaller [BF$_4$]$^-$ anion. There are a few causes of such behavior. One of them is that the smaller [BF$_4$]$^-$ anions are better packed in the RTIL liquid phase, which may limit their motion. Moreover, the [BF$_4$]$^-$ anions have a higher surface electrical charge density than [NTf$_2$]$^-$ and, thus, lower ionic mobility. However, it should be noted that there is an opposite effect, namely, the larger size of the $[NTf_2]^-$ anion restricts the dynamic movement and lowers σ [92]. However, as can be seen from Fig. 19b, which shows the temperature dependence of electrical conductivity of bulk RTILs, at high temperatures RTILs with a smaller $[BF_4]^-$ anion demonstrate higher conductivity than RTILs with a larger $[NTf_2]^-$ anion. For $[EMIM]^+$-based RTILs this temperature is lower than that for $[OMIM]^+$-based RTILs. Such tendency in the temperature dependence of electrical conductivity is in agreement with the experimental data obtained for [BMIM][BF_4] and [BMIM][NTf_2] [73]. This conductivity behavior can be explained by the fact that the interaction between the cations and anions is weakened resulting in a larger fraction of free ions, with the size and shape of anions beginning to play a more important role in determining the magnitude of electrical conductivity at a higher Temperature than under ambient conditions.

4. Conclusions

The structural, dynamic and electrical properties of four room-temperature ionic liquids differing in the alkyl chain length and anions in a nanoconfinement of porous carbons with different pore sizes have been studied by means of molecular dynamics simulations. The energetic characteristics obtained by using quantum-chemical calculations indicate weakening of the interactions between the ions with an increase in the cation and anion size. The density profiles for the confined RTILs show local maxima near the walls with higher density of the ions at the interface due to a strong electrostatic interaction between the RTIL and the charged graphene sheets. In the pores with $h \geq 2$ nm there are clearly observed adsorption double-layers near the walls. But in the narrow pores of 1 and 1.5 nm in width, we cannot distinguish a double electrical layer at each wall separately, since there is practically no free space for the anions due to the rather large size of the cations. Compared with other RTILs, the difference between the fractions of cations and anions for [OMIM][NTf_2] is maximal in the region of narrow pores. The lack of sufficient free space and at the same time the need to compensate for the strong electrostatic repulsion between the $[OMIM]^+$-cations result in the predominance of the *trans*-configuration of $[NTf_2]^-$ anions. Our results predict that the self-diffusion coefficients of the ions in ionic liquids are of the order of 10^{-11} m^2/s in the bulk phase at 300 K, which is in agreement with the literature data. The summed values of the cationic and anionic self-diffusion coefficients for the RTILs increase as follows: D[OMIM][BF_4] < D[EMIM][BF_4] < D[OMIM][NTf_2] < D[EMIM][NTf_2]. The presence of charged graphene surfaces complicates the RTIL dynamic behavior of ions in a confinement caused by preferred orientations and reduced mobility. The discrepancy in the MSD value in different directions caused by the layering structure of the RTILs inside the pores decreases with an increase in the pore width. The heterogeneous dynamics of the RTILs is characterized by slowing down of the ions in the layers closest to the walls and the dynamics of the ions in the center of the pore similarly to the dynamics in the bulk. The temperature influence on the dynamics of the ions in a 2-nm-wide pore, as well as in the bulk phase, was analyzed by studying [OMIM][NTf_2] and [EMIM][NTf_2] as an example. It was found that at a high temperature the diffusive regime for the z-component of MSD is achieved for all the ions and the cations become capable not only of oscillating near the pore walls in the Z direction but also of moving from one charged surface the other. As the temperature rises to 500 K, the self-diffusion coefficients increase by two orders of magnitude. The RTILs with a longer alkyl chain ([OMIM][BF_4] and [OMIM][NTf_2]) show lower electrical conductivities than [EMIM][BF_4] and [EMIM][NTf_2], regardless of temperature. The same effect is observed for the RTILs with different anions at a high temperature. But for temperatures below 450 K, the RTIL with a larger $[NTf_2]^-$ anion has higher conductivity than the RTIL with a smaller $[BF_4]^-$ anion, which is in agreement with the experimentally observed trend for the other RTILs.

CRediT authorship contribution statement

Darya Gurina: Conceptualization, Investigation, Methodology, Validation. Visualization, Writing – original draft. **Ekaterina Odintsova:** Investigation, Validation. Visualization, Writing – original draft. **Mikhail Krestianinov:** Investigation, Methodology, Validation. Visualization, Writing – original draft. **Yury Budkov:** Conceptualization, Project administration, Resources, Supervision, Methodology, Validation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2023.122961.

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